Tropical cyclone - a giant heat engine

Y12 (2012) — English translation

This problem considers a simplified model of a tropical cyclone. In this model, the cyclone is considered as a giant heat engine operating according to the Carnot cycle.

Rice. 1. Diagram of air mass flows

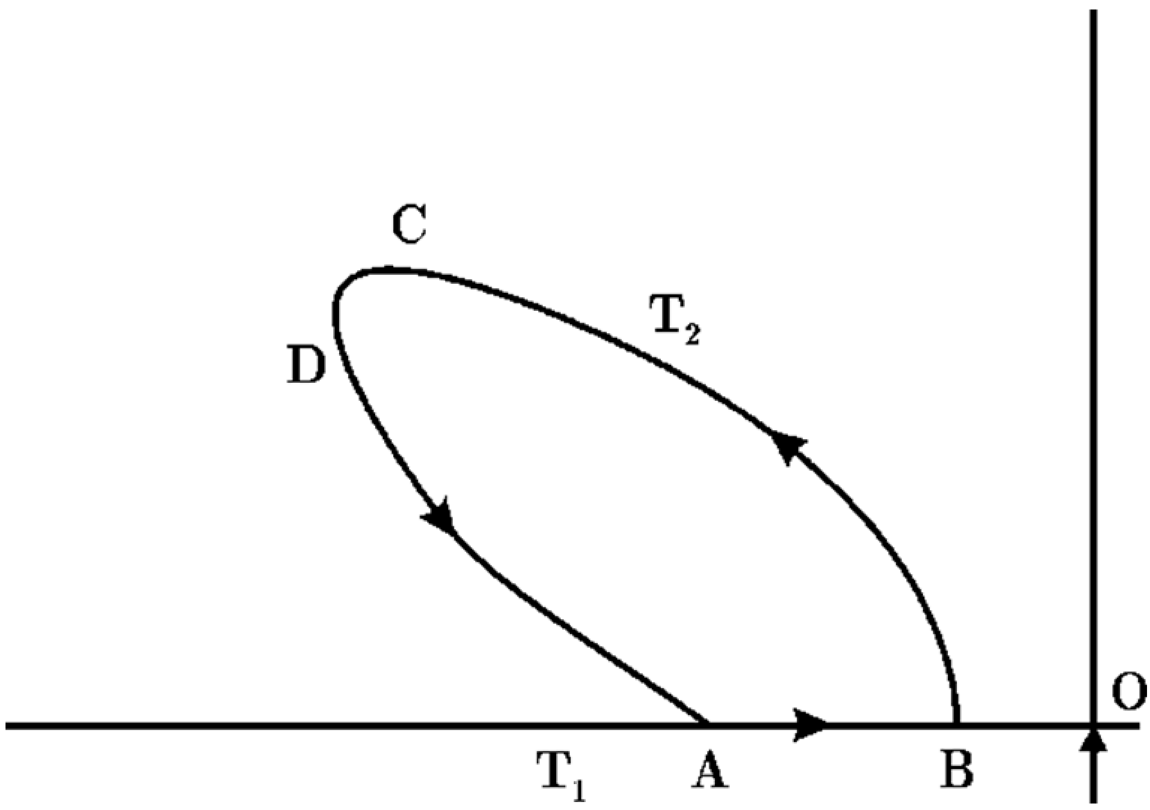

Fig. 1.

Rice. 2. Carnot cycle

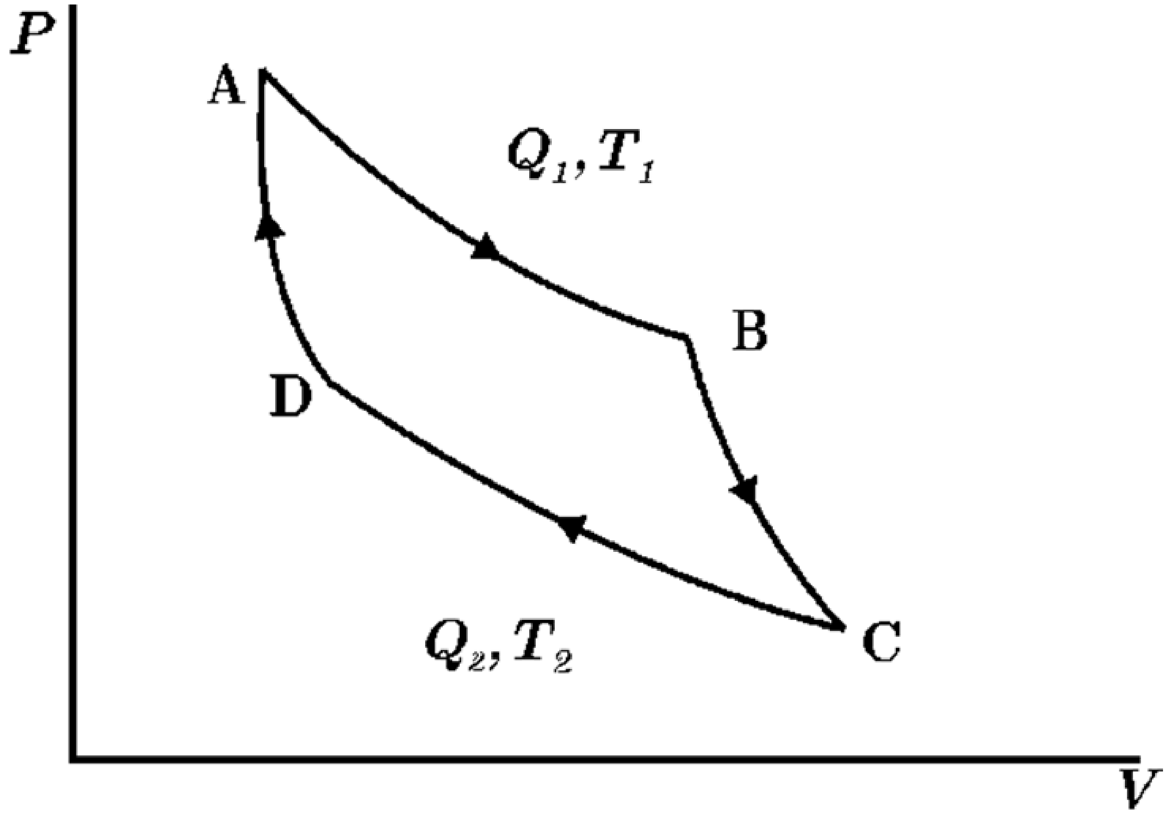

Fig. 2.

A mass of dry air Δm flows along the ocean surface from a high pressure area (A) to a low pressure area (B) close to the center of the storm O (Fig. 1), capturing moisture (water vapor) along the way. Together with the vapors, heat Q_1 is taken from the ocean. Air masses saturated with water vapor rise into the troposphere (section BC). In the troposphere (section CD) they are discharged as a tropical shower, giving off energy Q_2.

A1	Using the first law of thermodynamics, find an expression for Q_1 (see Fig. 2) in terms of L - the specific heat of evaporation, Δq - the moisture mass of the trapped air mass Δm on the way from A to B P_A and P_B - pressure in point A and point B respectively. Use the universal gas constant R, sea surface temperature T_1, molar mass M of air.	**0**

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A2	Find the total work done in one Carnot cycle by a heat engine on a mass of air Δm.	**0**

A3	Assuming that the work done by the heat engine goes entirely to increasing the kinetic energy of the air mass, obtain an expression for the speed v_B of the air in the central part of the cyclone (the “eye” of the cyclone).	**0**

A4	Assuming $v_A = 0$, find the numerical value of v_B using the following data: $T_1 = 273\,\text{К}$, $T_2 = 213\,\text{К}$, $R = 8.31\,\text{Дж}/(\text{моль}\cdot\text{К})$, relative air humidity at B $\varphi = 75\%$, at $T = T_1$ saturated vapor pressure $P_H = 42$ мбар, $P_A = 1000$ мбар, $P_B = 950$ мбар, average molar mass of air $M = 29\cdot 10^{-3}\,\text{кг}\cdot\text{моль}^{-1}$, molar mass of water $M_\text{в} = 18\cdot 10^{-3}\,\text{кг}\cdot\text{моль}^{-1}$, specific heat of evaporation of water $L = 2.26\cdot 10^6\,\text{Дж}\cdot\text{кг}^{-1}$, density of dry air $\rho = 1.3\,\text{кг}/\text{м}^3$ (1 бар $= 10^5$ Па).	**0**

A5	Assuming that the azimuthal velocity v and the radial distance r from the center of the cyclone are related by $v\sqrt{r} = \text{const}$, and that the velocity v_A at the outer edge of the cyclone is $10\,\text{м}/\text{с}$, calculate the effective radius of the cyclone (assume $r_B = 10$ км).	**0**

Source: https://pho.rs/p/3971